3K Compound Planetary Reduction Gearbox With Non-backlash Mechanism

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***Abstract*—It is a challenge to improve accuracy while maintaining high efficiency in a 3K compound planetary reduction gearbox with a high reduction ratio. In this paper, we focus on the traction drive with the same transmission structure and propose a hybrid of traction drive and a 3K compound planetary reduction gearbox. As a result, besides conventional high torque transmission and high efficiency, minute torque transmission is also possible. In this paper, the effectiveness of the proposed mechanism was confirmed by experimenting forward drive efficiency and backlash.**

***Index Terms*—3K Compound planetary reduction gearbox, Traction drive**

I. Introduction

Industrial robots are now indispensable at various production sites including the automobile and electronic equipment manufacturing industries, and the number of operating robots is increasing in emerging countries coupled with AI and automation needs [1]. Many industrial and humanoid robots use speed reducers as actuators [2].

Currently, various mechanisms such as planetary gearboxes using involute teeth as rotary actuators, wave gears known as harmonic drives, and cycloid pin wheel reducer are used to meet the demands of robots. [3], [4]. The planetary reduction gearbox has the advantages of large torque capacity, high speed reduction, high efficiency [5], but it has the disadvantage of increasing the backlash because it is designed to have a gap in the meshing part. Backlash is one of the most important nonlinearities that can limit the performance of speed and position control in robotics [6]. For high-precision positioning, the current numerical control algorithm uses the backlash correction method in many situations [7], [8]. On the other hand, there is a mechanism for suppressing backlash of reduction gears. One of them is called a wave gear reducer. The wave gear reducer has no backlash or is tiny, so it is considered suitable for precision positioning. However, wave gear reducers are very expensive due to the special metal and structure. In terms of efficiency, the reduction ratio is about 100, and the efficiency remains at 60-80% [4].

Matsuki et al. [4] proved that the back drivability increased with the increase in forward drive efficiency, and derived the efficiency equation dynamically, achieving a high efficiency of 89% in the actual machine. However, even if high efficiency is achieved, high accuracy cannot be expected as long as it is a planetary mechanism with backlash.

Xiaolan [9] proposed a traction drive that uses rollers instead of gears as a high-precision reduction mechanism. The structure is a planetary mechanism. According to Xiaolan [9], there is no backlash because of the transmission due to the rolling contact between the rollers, there is no backlash, and very high accuracy can be achieved, but the torque capacity will be small. In this paper, we propose a new reduction gear with high efficiency, high accuracy, and high back drivability by realizing the hybrid structure of traction drive and 3K compound planetary reduction gearbox with the same structure.

II. 3K compound planetary gear mechanizm

Fig.1 shows the structure of a 3K compound planetary mechanism when the number of planet gears $N = 3$. The planet gear P1 revolves around the sun gear S along the ring gear R1, as shown in Figs 1(a),(b). Also, the first ring gear R1 is fixed, the first planet gear P1 and the second planet gear P2 are rotated together using the carrier C, and the second ring gear R2 is output. As a result, a large reduction ratio can be obtained. In this complicated structure, if the number of teeth of each gear is expressed as z_i, $i \in \{s, p1, p2, r1, r2\}$, the transmission direction of the output with respect to the input is $z_{p1} > z_{p2}$ and $z_{r1} > z_{r2}$, opposite when $z_{p1} < z_{p2}$ and $z_{r1} < z_{r2}$ [4].

The forward drive efficiency formula of the 3K compound planetary reduction gearbox is obtained from the following conditional formula [4].

- Case of $I_2 < 1$. ($z_{r1} > z_{r2}$ and $z_{p1} > z_{p2}$)

$$\eta = \frac{\tau_{out}\omega_{r2}}{\tau_{in}\omega_s} = \frac{(1+\eta_a\eta_b I_1)(1-I_2)}{(1+I_1)(1-\eta_b\eta_c I_2)} \tag{1}$$

$$\eta' = \frac{\tau_{out}\omega_s}{\tau_{in}\omega_{r2}} = \frac{(1+I_1)\eta_a(\eta_b\eta_c - I_2)}{\eta_c(\eta_a\eta_b + I_1)(1-I_2)} \tag{2}$$

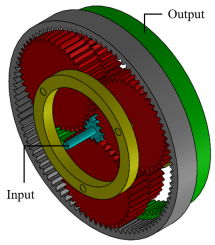
- Case of $I_2 > 1$. ($z_{r1} < z_{r2}$ and $z_{p1} < z_{p2}$)

$$\eta = \frac{\eta_c(\eta_b + \eta_a I_1)(1-I_2)}{(1+I_1)(\eta_b\eta_c - I_2)} \tag{3}$$

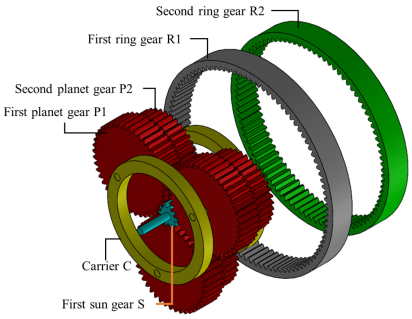
$$\eta' = \frac{(1+I_1)\eta_a(1-\eta_b\eta_c I_2)}{(\eta_a + \eta_b I_1)(1-I_2)} \tag{4}$$

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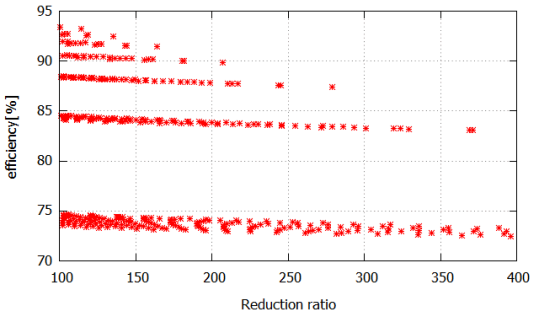


(a) An assemble model.

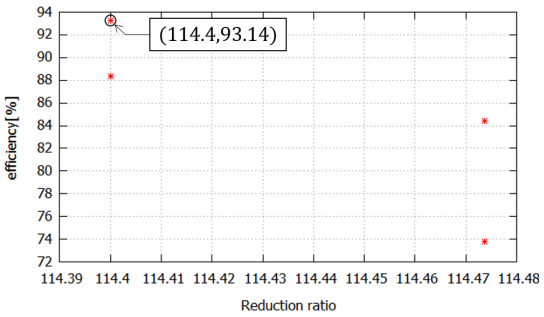


(b) A disassemble model.

Fig. 1. 3K compound planetary reduction gearbox mechanism.



(a) Reduction ratio of 100–400.



(b) Reduction ratio of 114–115.

Fig. 2. Forward driving efficiency distribution with reduction ratio.

Here, η is forward driving efficiency, and η' is reverse driving efficiency. I_1, I_2 and the relationship between angular velocity of input ω_s and output ω_{r2} are given by the following equations.

$$I_1 = \frac{r_{r1} r_{p11}}{r_s r_{p12}} = \frac{z_{r1}}{z_s} \tag{5}$$

$$I_2 = \frac{r_{r1} r_{p2}}{r_{r2} r_{p12}} = \frac{z_{r1} z_{p2}}{z_{r2} z_{p1}} \tag{6}$$

$$\omega_{r2} = \frac{1 - I_2}{1 + I_1} \omega_s = G_g \omega_s \tag{7}$$

$$\eta_i = 1 - \mu\pi \left(\frac{1}{z_{i1}} + \mathrm{sgn}_i \frac{1}{z_{i2}} \right), \; i \in \{a, b, c\} \tag{8}$$

where r_i, $i \in \{s, p2, r1, r2\}$ is the radii of the pitch circle of each gear, r_{p1i}, $i \in \{1, 2\}$ is the radius of the pitch circle of the first planet gear when engaging the S and R1, respectively. $1/G_g$ is the reduction ratio. If $I_2 = 1$, the reduction ratio becomes 0 and rotation is not possible. η_i, $i \in \{a, b, c\}$ is the basic efficiency between S–P1, P1–R1 and P2–R2, respectively. The function $\mathrm{sgn_i}$ is positive when the external gears mesh with each other, and is negative when the external gears and the internal gears mesh with each other. Therefore, it is positive between S–P1, negative between P1–R1 and P2–R2. Hence, the sign of function is $\mathrm{sgn_a} = 1$, $\mathrm{sgn_b} = \mathrm{sgn_c} = -1$. Since this basic driving efficiency is determined by the number of teeth and the shift coefficient, the efficiency can be maximized by the number of teeth and the shift coefficient [4].

A. *Maximize efficiency*

As a forward drive efficiency optimization problem, the variable parameters and constraints to be determined are:

$$\text{maximize } \eta(z_s, z_{p1}, z_{p2}, z_{r1}, z_{r2}, x_s, x_{p1}, x_{p2}, x_{r1}, x_{r2}) \tag{9}$$

$$\text{subject to} - 2 \leq x_s, x_{p1}, x_{p2}, x_{r1}, x_{r2} \leq 2 \tag{10}$$

$$r_a = r_b = r_c \tag{11}$$

Where, (10) is a general restriction of the profile shift coefficient, (11) is a restriction due to the coaxial condition that exists as a constituent condition of the 3K planetary mechanism, while the carrier is coaxial with P1 and P2. Note that r_i, $i \in \{a, b, c\}$ is the distance of center between S–P1, P1–R1 and P2–R2, respectively. The relationships of profile coefficient are given by the following formula:

$$x_s + x_{p1} = \frac{z_s + z_{p1}}{2 \tan \alpha} (\mathrm{inv}\alpha_{wa} - \mathrm{inv}\alpha) \tag{12}$$

$$x_{r1} - x_{p1} = \frac{z_{r1} - z_{p1}}{2 \tan \alpha} (\mathrm{inv}\alpha_{wb} - \mathrm{inv}\alpha) \tag{13}$$

$$x_{r2} - x_{p2} = \frac{z_{r2} - z_{p2}}{2 \tan \alpha} (\mathrm{inv}\alpha_{wc} - \mathrm{inv}\alpha) \tag{14}$$

where $\mathrm{inv}\alpha$ is an involute function defined by $\mathrm{inv}\alpha = \tan\alpha - \alpha$. Here, α is the basic pressure angle. The α_{wi}, $i \in \{a, b, c\}$ is the working pressure angle between S–P1, P1–R1, and P2–

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TABLE I
MAXIMUM EFFICIENCY PARAMETERS IN FIG. 2(B).

Symbol	Contents	Value
η	Efficiency	93.14
m	Module	0.8
$1/G_g$	Reciprocal of reduction ratio	114.4
z_s	Number of first sun gear	10
z_{p1}	Number of first planet gear	44
z_{p2}	Number of second planet gear	37
z_{r1}	Number of first ring gear	98
z_{r2}	Number of second ring gear	91
x_s	Profile-shift coefficient of first sun gear	0.4010
x_{p1}	Profile-shift coefficient of first planet gear	0.79896
x_{p2}	Profile-shift coefficient of second planet gear	0.75963
x_{r1}	Profile-shift coefficient of first ring gear	2.0
x_{r2}	Profile-shift coefficient of second ring gear	1.95889

R2, respectively, and is given by the following equation.

$$\cos\alpha_{wa} = \frac{d_{bs} + d_{bp1}}{2r_a} = \frac{(m_s z_s + m_{p1} z_{p1})\cos\alpha}{2r_a} \tag{15}$$

$$\cos\alpha_{wb} = \frac{d_{br1} - d_{bp1}}{2r_b} = \frac{(m_{r1} z_{r1} - m_{p1} z_{p1})\cos\alpha}{2r_b} \tag{16}$$

$$\cos\alpha_{wc} = \frac{d_{br2} - d_{bp2}}{2r_c} = \frac{(m_{r2} z_{r2} - m_{p2} z_{p2})\cos\alpha}{2r_c}. \tag{17}$$

Here, d_{bi}, $i \in \{s, p1, p2, r1, r2\}$ is a basic circle diameter of each gear, and is expressed as $d_{bi} = z_i m_i \cos\alpha$, where m_i is a module. Note that $m_s = m_{p1} = m_{r1}$ and $m_{p2} = m_{r2}$. Under these constraints, efficiency was maximized by optimizing the number of teeth z and the profile shift coefficient x. Fig. 2(a) shows the distribution of the maximized efficiency versus the reduction ratio. From the efficiency distribution chart Fig. 2(b), the most efficient reduction ratio of 114.4 among those with a reduction ratio exceeding 100 was adopted. The optimization was performed using the quasi-Newton method. The optimized parameters are shown in the table I.

III. TRACTION DRIVE

There is a traction drive as a reduction gear with high accuracy. The traction drive uses contact between rollers as a transmission force instead of gears. Since it has the same planetary mechanism as the planetary reduction gearbox, the speed ratio is also determined by the ratio of the diameters of roller. This paper describes a method for connecting the 3K compound planetary mechanism and the planetary traction mechanism in parallel.

A. Transfer method

Traction drive is a method of transmitting force through a lubricant film named as traction oil from one rotating body to the other using rollers that are pressed against each other. As shown in Fig. 3, in the traction drive, power is transmitted from the drive side to the driven side by a traction oil film interposed between two pressure-contacted rollers. There is a slight relative speed difference between the rollers, and a tangential force F_t is generated because the oil film is sheared. The tangential force F_t is given by

$$F_t = \mu_t F_n. \tag{18}$$

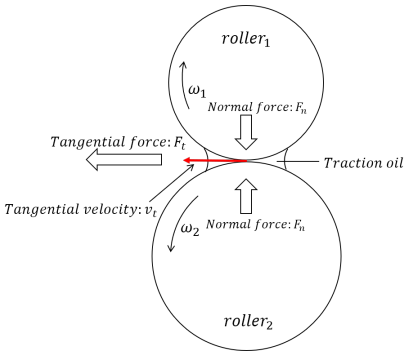


Fig. 3. Basic principles of the traction drive.

where μ_t is called the traction coefficient, and F_n is the pressure force between the rollers. Since the two rollers are rolling and transmitting, the rotation accuracy is high and the backlash is zero.

IV. THE PROPOSED MECHANISM

We propose a hybrid of a traction drive and a 3K compound planetary reduction gearbox that maximizes efficiency. As a result, both high torque transmission by the gears and minute torque transmission by the rollers can be realized at the same time, so transmission is possible even between the backlashes, and the positioning accuracy increases dramatically. The hybrid mechanism is shown in Fig. 4. The names in the traction drive are represented as the sun roller $\mathrm{S_r}$, the first planet roller $\mathrm{P1_r}$, the second planet roller $\mathrm{P2_r}$, the first ring $\mathrm{R1_r}$, and the second ring $\mathrm{R2_r}$, respectively.

Also, connecting 3K compound planetary reduction gearbox and traction drive increases the loss, so there is no reason to maximize efficiency. Therefore, it is necessary to design a minimum loss in the traction drive.

A. Design a minimum loss

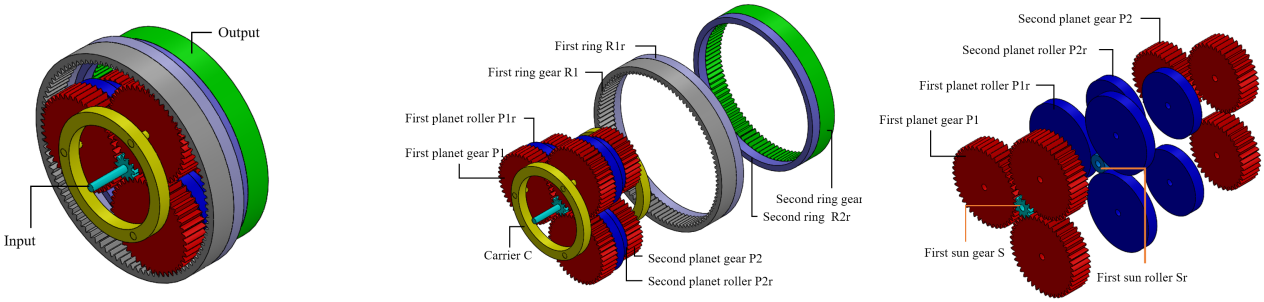
The problem to be considered in the friction transmission system is the loss caused by slipping. In Fig. 3, the drive side is $\mathrm{roller_1}$ and the driven side is $\mathrm{roller_2}$. A state where the tangential velocity difference between the driving side and the driven side is zero is called rolling contact, and when there is slippage, it is called sliding contact. Since the loss is smaller in the rolling contact than in the sliding contact, it is necessary to install the roller to be in rolling contact. If the angular velocity ratio of the gear coincides with the diameter ratio of the roller, the tangential speeds on the drive side and the driven side are equal.

The speed relation satisfying the rolling contact is expressed here using the working pitch circle radius by the following formula (19). The working pitch circle radius r_{wi} is given by Eq.(20)

$$\frac{\omega_2}{\omega_1} = \frac{r_{w1}}{r_{w2}}, \tag{19}$$

$$r_{wi} = \frac{r_{bi}}{\cos\alpha_{wj}},\ i \in \{s, p1, p2, r1, r2\},\ j \in \{a, b, c\}. \tag{20}$$

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(a) An assemble model. (b) A disassemble model. (c) A disassemble carrier.

Fig. 4. The proposed mechanism.

TABLE II
DIAMETER OF EACH ROLLER.

Symbol	Contents	Value
d_{sr}	Diameter of first sun roller	$8.315[\mathrm{mm}]$
d_{p1r}	Diameter of first planet roller	$36.58[\mathrm{mm}]$
d_{p2r}	Diameter of second planet roller	$30.76[\mathrm{mm}]$
d_{r1r}	Diameter of first ring roller	$81.48[\mathrm{mm}]$
d_{r2r}	Diameter of second ring roller	$75.66[\mathrm{mm}]$

In the above, r_{w1} is a working pitch circle radius on the driving side, and r_{w2} is a working pitch circle radius on the driven side. This formula can be easily satisfied if the rollers are in contact with each other, but the 3K compound planetary mechanism requires constraints. There are two working pitch circle radii only in the first planet gear P1, as shown in equations (5) and (6). The reason is indicated in Fig. 5. The working pressure angle for the first sun gear α_{wa} is different from the working pressure angle for the first ring gear α_{wb}. Sometimes, the working pitch circle radius r_{wp1} changes as in the expression (20). Therefore, when the working pressure angle is different, the mechanism becomes complicated because there are two diameters of roller in the first planet gear. Hence, the working pressure angles α_{wa} and α_{wb} must be equal. The coaxial condition (11) must be satisfied in order for (15) and (16) to be equal. This time, when maximizing efficiency, it is set to satisfy the constraints, so in the first planet gear there is only one working pitch circle diameter. Consequently, the roller diameter d_{ir} is the working pitch circle diameter and is shown in Table II.

V. EXPERIMENT

Two reduction gearboxes with the same number of teeth are prepared for maximum efficiency, with only one gear attached and the other attached with a roller. Hereinafter, in these two reduction gearboxes, a mechanism using only gears is referred to as an "existing mechanism", and a mechanism using gears and rollers is referred to as a "proposed mechanism". For the existing mechanism and the proposed mechanism, forward drive efficiency, backlash, were measured.

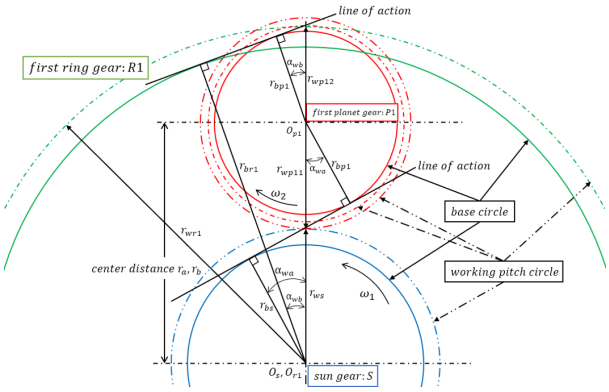


Fig. 5. Gear transmission relationship diagram

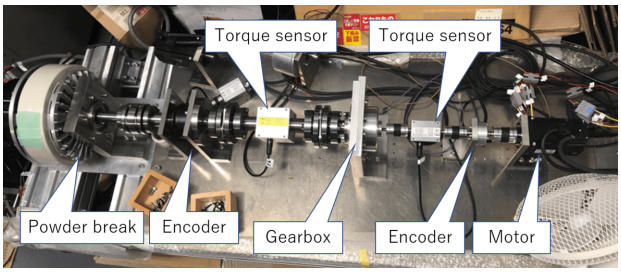
A. Efficiency measurement

The experimental environment shown in Fig. 6(a) was set up. We used a motor (SGMJV-04A YASKAWA), an encoder (MTL MG30) with encoder resolution 300 $\mathrm{P/R}$, and a torque sensor (UTM2-2Nm) as input shafts. On the output side, load torque (ZKB-20HBN MITUBISHI), an encoder (MTL MEH-60) with encoder resolution 108,000 $\mathrm{P/R}$ and torque sensor (UTM2-200Nm) were used. The motor was speed controlled and digitally controlled using PR-Expert4. The control cycle is 100 μs. We measured efficiency under the following conditions. Input was from 500 rpm to 3000 rpm in 250 rpm increments and load torque was from 10 $\mathrm{N.m}$ to 50 $\mathrm{N.m}$ in 10 $\mathrm{N.m}$ increments. Calculations were made using the following equations from the values of the encoder and torque sensor on the input side and output side, respectively.

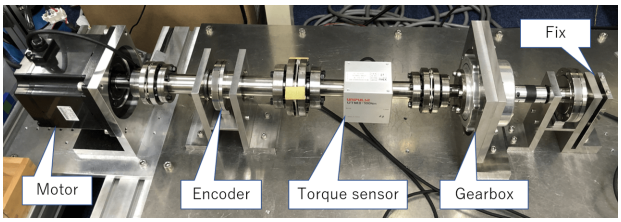
$$\eta_{exp} = \frac{1/T\sum \omega_{out}\tau_{out}}{1/T\sum \omega_{in}\tau_{in}} \times 100\,[\%]. \tag{21}$$

The experimental results are shown in Figs. 7(a) and 7(b). In the results, the x-axis is the input angular velocity (rpm), and the y-axis is the load torque $(\mathrm{N.m})$ and using a color map of efficiency. As a result, the average efficiency of the existing mechanism was 88.519% and the proposed mechanism was 87.118%. Also, compared to the existing mechanism, the

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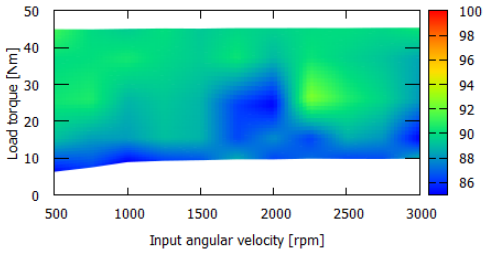


(a) Testbench for measurement efficiency.

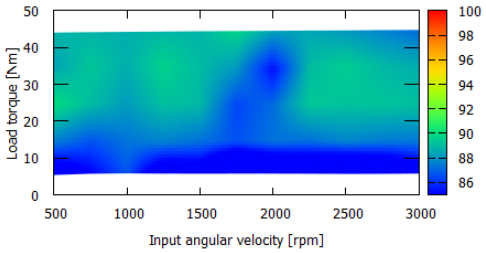


(b) Testbench for measurment backlash.

Fig. 6. Experiment equipment.



(a) Exsiting mechanism.



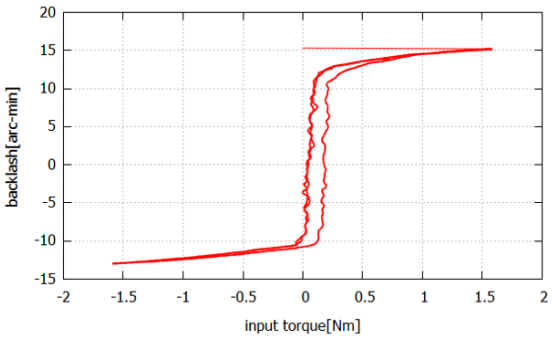
(b) Proposed mechanism.

Fig. 7. Results of measurement efficiency with distribution.

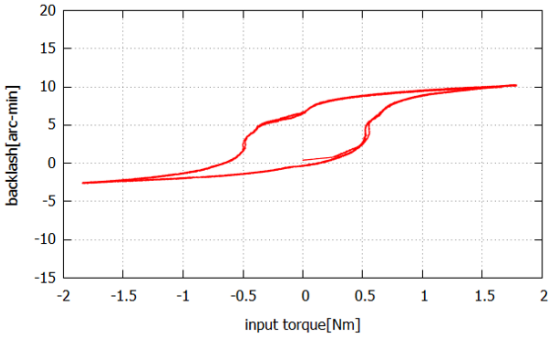
proposed mechanism appears to be slightly lighter overall. This is thought to be caused by the loss of the roller, but the difference is only about 1.401%, so it can be seen that the loss is considerably suppressed.

B. Backlash measurement

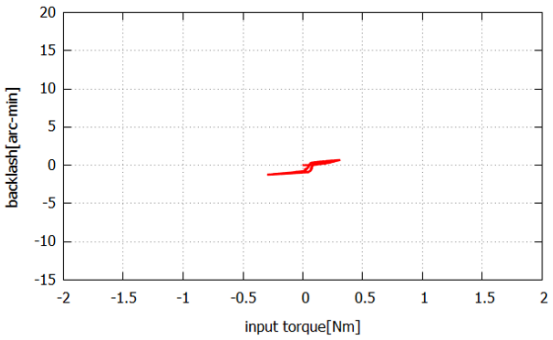
The experimental environment shown in the Fig. 6(b) was set up. The input shaft was fixed. A motor (SGMPS-15ACA61-E) was used on the output side. In addition, an encoder with encoder resolution 11,840,000 $\mathrm{P/R}$ was installed for input and output, respectively. First, the upper limit torque



(a) Exsiting mechanism ($\pm 2\mathrm{N.m}$).



(b) Proposed mechanism ($\pm 2\mathrm{N.m}$).



(c) Proposed mechanism ($\pm 0.5\mathrm{N.m}$).

Fig. 8. Result of measurement backlash with hysteresis curve.

was set, and the backlash was measured by rotating gradually in the positive direction at a constant speed, and inverting the direction of rotation when approaching the upper limit torque. The backlash calculation formula can be expressed as an angular transmission error θ_{error}. It is measured in minutes.

$$\theta_{error} = (\theta_{out} - G_g \theta_{in}) \times 60\,[\mathrm{min}] \tag{22}$$

The experimental results are shown in Figs. 8(a) 8(b) and 8(c) and the condition is different. The limit top torque was $\pm 2\,\mathrm{N.m}$ in (a) and (b), and $\pm 0.5\,\mathrm{N.m}$ in (c). Since the noise component of the torque sensor increases, the results of torque were smoothed through a moving average filter. Note that all moving average filters have 500 windows. In the results, the x-axis is the input torque $(\mathrm{N.m})$, and the y-axis is the angle transmission error (min). The backlash is the difference

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in angle transmission error when the input torque is zero. The experiment was performed three times in total, and the experimental result was one of them. The results show that the existing mechanism has about 23 min, but the proposed mechanism has become smaller, 7.87 min on average. These were the results of setting the upper limit torque at $\pm 2\,\mathrm{N.m}$, but the nonlinear element was seen near $\pm 0.5\mathrm{N.m}$ in Fig. 8(b). Therefore, the result of applying torque less than $\pm 0.5\mathrm{N.m}$, which is transmitted only by the roller, is shown in Fig. 8(c), and the backlash amount is tiny at 1.27 min on average. The rollers transmit by rolling contact, but there is a slight slip. Even if the backlash due to the gears no longer exists, the backlash still exists due to the rollers shown in Fig. 8(c).

VI. Conclusion and Future works

The maximum efficiency between the reduction ratios of 100-400 was obtained from the efficiency equation for the 3K compound planetary gear mechanism, and the theoretical efficiency of 93.14% was achieved at the reduction ratio of 114.4. Also, from the basic principle of traction drive, a roller is installed on the working pitch circle to meet the rolling contact conditions. Thus, loss is minimized. Each experimental result showed that the loss of efficiency in the hybrid mechanism had little effect. As future works, we want to create a dynamic model and combine the control model to show its usefulness from the viewpoint of control.

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